

Do **not** write solutions on this page.

Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 14]

Two athletes, Fiona and Lucy, compete in a 200 metres race along a straight track.

Fiona's velocity, in m s^{-1} , during the race can be modelled by $v(t) = \frac{8.14t}{\sqrt{t^2 + 0.2}}$, where $t \geq 0$. Time, t , is measured in seconds from when the race starts.

- (a) (i) Write down the value of $v(1)$.
- (ii) Find the time when Fiona's velocity is 5 m s^{-1} . [3]
- (b) Find the time when Fiona's acceleration is 4 m s^{-2} . [2]
- (c) (i) Write down the limit of $v(t)$ as t approaches infinity.
- (ii) State a reason why the value in part (c)(i) is not valid in the context of this question. [3]

Lucy's velocity, in m s^{-1} , during the race can be modelled by $w(t) = \frac{8t}{\sqrt{t^2 + 0.3}}$, where $t \geq 0$.

Fiona completes the race and crosses the finishing line in front of Lucy.

- (d) Find the distance Lucy is from the finishing line when Fiona completes the 200 metres. [6]

